



Bridging AI and Emotion: Enhanced Models for Personal Finance Manager Applications

Samir Bader¹ and Ramzi A. Haraty²

¹ Department of Computer Science and Mathematics, Lebanese American University, Beirut, Lebanon

Received 6 June 2024, Revised 22 March 2025, Accepted 22 March 2025

Abstract: paper focuses on creating and deploying an innovative Financial Advisor Application that would revolutionize conventional financial planning methods with the help of state-of-the-art artificial intelligence tools and effective financial services. Indeed, the essence of this application is user-oriented and focuses on creating more ways to engage the user and help make the right decisions by redesigning the interface and providing individualized financial guidance. The application combines different aspects of financial management, including budgeting, transaction tracking, and goal setting, but also sees novelties as an artificial intelligence function that detects anomalies in transactions and the merchant offers explicitly created for clients. The technical structure involves access to data, the description of each module, and the use of AI that provides anticipatory analysis and tools for advanced financial planning. Business testing and vulnerability assessments were performed proving the business value and reliability of the application; thus, avoiding data leakage and modularity viruses. The work assesses the effects on the users' satisfaction when provided with such technologies and their ability to change their financial management behaviour; therefore, providing an exhaustive scenario of the credentials of AI for improving the efficacy of financial advisory. The conclusions released in the paper should advance knowledge of financial technology as it demonstrates how the intelligent systems used in a presented application can lead to noteworthy improvements for both users and financial companies.

Keywords: AI, Anomaly Detection, Financial Planning, Fintech, and Sentiment Analysis

1. INTRODUCTION

The Financial Advisor Application is a perfect example of the synergy of highly developed IT solutions and profound financial knowledge in the constantly evolving field of fintech. This application revolutionizes personal finance management by utilizing advanced AI along with a plethora of financial products and services that are unique to the clients. Unlike traditional instruments that work independently of one another, this platform offers a global picture of a person's financial status and provides recommendations and solutions.

Due to the advanced semantic AI, the Financial Advisor Application is proficient in identifying users' emotions regarding financial activities and textual information. This advanced feature allows the application to provide more personalized financial advice and emotional support, making the financial management process more accessible and sensitive. As the program determines the user's emotional state when making financial decisions, the application offers recommendations that match the feelings and objectives of the user; thus, increasing the interest and satisfaction of

users.

Also, constant transactions are controlled by the application through highly developed anomaly detection algorithms. This security feature quickly detects any suspicious activity associated with fraudulent activities; thus, protecting users' financial resources. The anomaly detection ability protects the money and makes users aware of standard spending patterns to avoid risking their money.

Apart from these features, the application also customizes the shopping experience by identifying the merchants with whom the users have transactions. This feature provides a user-friendly experience and can save the most money by recommending discounts and promotions from associated sellers. As a result, the Financial Advisor Application dictates that users get only the best deals by considering past purchases and preferences, promoting financial optimality and customer happiness. It questions the conventional business models of financial consulting and modernizes the principles of the personal finance industry. It gives an insight into the prevailing position of the application of AI in the financial sector, which opens up

future financial decision-making as informative, inclusive, and innovative. When reflecting on the possibilities and features of the Financial Advisor Application, it is crucial to realize that this is not merely an application but a financial assistant that helps users in their economic endeavors with accuracy and wisdom.

A. Problem Statement

This desire for more effective tools to extend financial consultancy with the help of the internet and mobile applications highlights the necessity of stable methods to develop individual, particular approaches for every user. Most contemporary approaches to managing financial and accounting activity are based on fixed structures that ignore essential data, such as general user behavior and attitude needed to define what kind of recommendations the administrators should give. The problem is that these applications become mere carriers of simple data transfers and cannot analyze user data such as frequency, mood, and preferences crucial to providing timely financial assistance [1-3].

Today's financial platforms can process structured data effectively, they cannot extract as much value as from the unstructured data that stems from inputs about the users and descriptions of the transactions [1]. This can lead to recommending other products or services that the user does not need or cannot handle emotionally; thus, limiting the possibility of positive interaction and satisfaction. Specifically, it has been found that studies on concrete applications of the soft computing paradigms, like the applications with semantic analysis or the applications that include adaptive learning mechanisms, help solve the problems of financial advising. However, the practical application of these complex methods requires effort.

Regarding the above issue, an effective approach for improvement is to combine TensorFlow and Keras with a Python environment. This integration will help the application learn the users' activities and preferences by integrating semantic AI and M-learning solutions. So, achieving these objectives will facilitate the receipt and analysis of transactional data such as the number of transactions, the frequency of transactions, the currency, and any other quantitative indicators. It will also process inputs that contain quality or quantity data, such as the quantity of product, the quality of material, and so on. The knowledge of the users' requirements will be constructed by applying artificial neural networks in this application. This functionality can give general information concerning financial conduct and give particular recommendations depending on the user's data, significantly increasing user satisfaction.

This study seeks to fill the gap between conventional financial advisory platforms and advanced AI-based solutions by developing a financial advisor application that analyzes, comprehends, and responds to each user's economic story. This approach solves the crucial problem of improving the quality and effectiveness of digital services' financial recommendations, which raises users' confidence in financial

planning applications.

B. Scope of the Work

This work is wholly devoted to the user. Our vision is to create a highly sophisticated Financial Advisor Application as a service and a sidekick in the financial life. This work involves leveraging the features of NET Core 6 to design an extremely robust and highly scalable application. We have integrated advanced features of AI in Python, which are generally used for deep learning, such as anomaly detection, consecutive transaction prediction, and semantic analysis [2]. By using this diverse approach, we aim to enhance a user's experience by providing timely and accurate information on managing money. This project has been developed with the needs of user experiences in mind.

It is also not simply a matter of incorporating the different and complex aspects of modern financial advisory services' features into the application's architecture. It is about developing a framework that is open and capable of changing according to the conditions of the field. Abstract classes define the application structure, and this decision makes it possible to add new features and change the product's behavior without affecting its essential characteristics by third parties.

Implementing AI and machine learning into the application is not just a technicality. It's a strategic decision to address three primary objectives: anomaly detection for better transaction security, predictive analysis for better financial decisions, and semantic analysis for better financial planning advice.

Among them, the most important is to develop and optimize deep learning models for detecting anomalies in financial transactions. Python and other necessary machine learning libraries will be used to create the application that will assist in identifying suspicious activity and increasing transaction security. We know how critical security and reliability are in financial applications and strive to provide you with a product you can rely on.

To develop predictive models for financial forecasting: The application will analyze historical financial data and the user's transaction history to predict specific financial outcomes. This will empower users to make informed and accurate economic decisions.

To incorporate semantic analysis for understanding user sentiment and behavior, the application will use NLP techniques to better understand user inputs and feedback and provide more accurate and practical financial planning recommendations based on the client's emotional status.

The research questions guiding the development of this application are:

- In what ways can deep learning models be used to solve the problem of identifying anomalous transac-
-

tions? NET Core 6 environment?

- What methods can be used to effectively forecast financial patterns and users' actions based on transactional data using Python-based AI tools?

C. Research Objectives

This work aims to bridge the gap between conventional financial advisory platforms and advanced AI-driven solutions by developing a sophisticated Financial Advisor Application. The key objectives of this research are:

- Enhancing Financial Security – Develop and optimize deep learning models for detecting anomalies in financial transactions, improving transaction security through AI-driven anomaly detection techniques.
- Predictive Financial Analysis – Create predictive models that analyze historical financial data and user transaction histories to forecast financial trends and empower users with informed decision-making.
- AI-Powered Financial Planning – Incorporate semantic analysis and natural language processing (NLP) to understand user sentiment and behavior, enabling the application to provide personalized financial advice based on users' emotional and financial conditions.

D. Research Contributions

This work contributes to the field of AI-driven financial advisory systems by:

- Developing a Scalable and Adaptive AI-Driven Financial Advisory Framework – The application architecture is designed with abstract classes, enabling adaptability and extensibility in financial advisory services.
- Integrating Advanced AI Techniques for Financial Advisory Services – By leveraging deep learning for anomaly detection, predictive analytics, and semantic analysis, this study enhances the accuracy and reliability of financial recommendations.
- Enhancing Transaction Security in Digital Financial Services – Implementing anomaly detection techniques helps identify suspicious activities, increasing users' confidence in financial applications.
- Improving User-Centric Financial Advisory Services – The application's NLP-based semantic analysis tailors financial advice to users' emotional and financial states, providing a more intuitive and personalized experience.

The rest of the paper is organized as follows: Section 2 provides the literature review. Section 3 describes the methodology of the work. Section 4 presents the experimental results and section 5 supplies a discussion of the results.

And finally, section 6 concludes the paper and provides future directions.

E. Selecting a Template (Heading 2)

First, confirm that you have the correct template for your paper size. This template has been tailored for output on the 21cm X 28cm Paper Size.

F. Maintaining the Integrity of the Specifications

The template is used to format your paper and style the text. All margins, column widths, line spaces, and text fonts are prescribed; please do not alter them. You may note peculiarities. For example, the head margin in this template measures proportionately more than is customary. This measurement and others are deliberate, using specifications that anticipate your paper as one part of the entire proceedings, and not as an independent document. Please do not revise any of the current designations.

2. LITERATURE REVIEW

The spread and development of fintech have fostered further development of new applications that would contribute to the simplification of financial transactions and increase the accuracy and personalization of the offered financial recommendations. Among these innovations, the Financial Advisor Application is one of the best integrated and sophisticated applications in the world of technology. This paper aims to provide a literature review to establish the essential technologies and critical applications that define this revolutionary application. This new financial advisory tool combines contemporary computational techniques with traditional but customized services.

A. Comprehensive Financial Overview

The application gives users a summary of the financial situation, combining the data from several accounts and the asset summary, liability, expenditure, and savings plan. Users can make sound financial decisions by offering precise and concise financial decisions since the application provides a clear and concise financial picture. This feature corresponds with Anderson, who states that data integration and visualization play a crucial role in improving user experience and their decision-making process [1].

B. Technological Base and Management Framework

As stated earlier, the basis of the described Financial Advisor Application is the use of .NET Core — a decision that signals the readiness to build solid and scalable apps [3]. NET Core is a highly adaptive, fast, and secure platform crucial for financial applications involving users' data. It ensures that complex capabilities are closely integrated and the application will not slow down or have issues with various loads. This is especially the case within the financial services industry, where the accuracy of data and its authenticity cannot be compromised.

It is integrated with NET Core, and the application utilizes Python's AI features, including TensorFlow and Keras. These libraries are among the most advanced tools

for deep learning, which can analyze enormous amounts of information successfully and search for valuable information. Consequently, the use of the two has become mixed up. NET Core for structure and Python for intelligent operations are the best pairs that bring high efficiency and intelligence to the Financial Advisor Application.

C. Future Directions and Research Opportunities

Thus, the application of AI in financial advisory services offers many potential research avenues, especially regarding the protection of users' data, the usage of responsible AI, and the suitability of AI in financial decision-making. Future research could focus on the effects of real-time data analysis on economic forecasting, the utilization of AI to improve financial awareness, and how best to increase the degree of customization of financial advice using more sophisticated machine learning algorithms.

Moreover, the application's architecture and features allow for integrating quantitative and qualitative sentiment analyses of financial statements. Thus, perpetuating the integration of these elements can lead to further efficiency and customization in economic consulting.

The Financial Advisor Application is a new generation of personal finance management tools that incorporate the best features. Based on NET Core 6 and AI technologies developed using Python, it will be possible to provide users with a powerful, safe, and highly customized experience [3]. The application stands out from other financial technology applications through its capacity to predict trends, identify deviations from standards, and analyze user sentiment. The continued advancement of the application will give a glimpse of the possibilities and difficulties of using AI in delivering effective and efficient financial services to users, propelling the development of better and more effective user-oriented financial technologies.

D. The Relationship between Deep Learning and Machine Learning

The programming tools most useful for machine learning in Python are TensorFlow and Keras, which can predict financial trends, detect anomalies, and analyze semantics [7-8]. These capabilities enable the application to detect transaction patterns, predict the users' future economic activities, and ascertain their needs. RNNs will do the data analysis, and insights and recommendations will be given with the help of data analysis. However, very subtle patterns in the data may not be easily detected.

Substantial evidence suggests that machine learning models should be incorporated into financial forecasting. For example, models help predict market trends and identify patterns humans may not notice. These capabilities are crucial for creating a predictive financial advisory setting that responds to the user's data shift.

E. Customization and Extensibility

When it comes to extensibility and customization, the application's architecture, especially the usage of abstract

classes, serves as a guideline [4]. This architectural decision enables third-party developers and financial institutions to extend the application's core functionalities to suit different needs while not risking the overall integrity of the application [5]. It is a strategic move that centralizes the needs of a global user base and the required financial regulations, languages, and operational customizations

This flexibility is essential to keeping the application up to date and effective in a rapidly evolving financial environment provided by abstract classes. This aligns with research by Tertilt and Scholz, who postulated that financial solutions must be flexible enough to suit users' needs.

F. Personalized Financial Advice

Personalization is the foundation of the Financial Advisor Application's proposition. As such, the application can provide recommendations consistent with the user's spending patterns and financial objectives through semantic analysis and predictive models. Whether it is about saving money, investing money, or even planning for the budget, the application is designed to help according to the particularity of the user. This level of personalization is accomplished through learning and development to ensure that the advice is helpful and efficient in the future.

G. AI-Driven Merchant Recommendations

For better user experience and more savings, the Financial Advisor Application also uses Artificial Intelligence to analyze users' transaction history and suggest merchant offers. This not only helps the consumer by making the experience more personal and tailored to their needs but also helps them save money with discounts and sales from the partnered vendors. The application also employs user data and artificial intelligence to provide users with the most enriching offers, thus giving value to their financial operations.

H. AI Predicting Trends

It has been proved that AI can perform very well in predictive solutions across different domains, particularly financial platforms. Historical data are significant because AI computers can find hidden patterns that human analysts may otherwise ignore. Deep learning and neural networks have been critical in helping identify these patterns. For instance, Rezaei et al. have described how deep learning algorithms can perform the stock market's future downtrends with a high accuracy level, which can be helpful for investment [6].

I. AI in Other Fields

Although AI is particularly incorporated in finance, it resonates in multiple industries. In healthcare, AI algorithms are used for outbreak prediction and deep learning is utilized to design individual patients' care plans [7]. In the same way, in retail, AI is used to keep track of the stock and make recommendations to customers [8]. These advancements also indicate that AI has the unique ability and capacity to tailor applications to revolutionize several industries and sectors [14-16].

J. AI and Sentiment Analysis

Considering the growing popularity of AI applications in various industries, it is paramount to explore customer sentiment analysis. It typically marks text mining techniques that seek to identify the tone of opinion users express. This technique finds broader usage in the financial world for ascertaining the market mood from news articles and tweets. A listing by Bollen et al. [9] revealed that the mood on Twitter could affect the movements of the stock market, making sentiment analysis crucial in market analysis.

K. Behavior Prediction

Customer behavior prediction using artificial intelligence includes identifying patterns and definitions of behavior to make assumptions about the likely behavior of customers in the foreseeable future. This is particularly helpful in marketing with personalized segments and customer maintenance programs. For instance, there are behavioral prediction models to determine which customers are most likely to churn, so businesses can take preventive measures [8]. This capability is also essential in processing financial expenditure since spending behaviors can provide valuable insights into financial management advice [10].

L. Forecasting

AI has enormously changed the different sectors through the foresighted analysis of the data collected. Some models, such as the LSTM and N-BEATS, have been used in finance studies, as shown in the time series forecasting. Such models can deal with intricate data patterns and offer robust prediction, which can help make decisions [11].

Therefore, there is enough ground for integrating machine learning models in financial forecasting. For instance, models aid in comprehending attitudes and climates that humans may not discern inherently. These capabilities are essential for constructing an accurate advisory financial environment that adapts to the alteration of data mentioned by users [12].

M. AI in Natural Language Processing

The developments in natural language processing (NLP) have contributed immensely to improving the capability and capacity of various forms of AI to interpret and create natural languages. These enhancements have ensured a better handling of user interactions and an excellent sentiment analysis. It is essential to discuss deep learning trends in NLP, the progress made, and the future that may lie ahead for these technologies, which was addressed by Young et al. in the perspective of 2018 [13].

N. AI in Financial Advisory and Planning

Recent research underscores the transformative impact of AI in financial advisory systems. For example, Wang et al. [14] explored the use of generative AI for dynamic portfolio optimization, demonstrating how reinforcement learning algorithms can provide real-time investment strategies. Similarly, Patel and Kumar [15] investigated AI-driven

budget planning tools that leverage predictive analytics to anticipate users' future financial needs based on historical data, showcasing improved financial decision-making outcomes.

AI's ability to detect anomalies in financial transactions has also been a focal point. Lee et al. [16] developed a machine learning framework for fraud detection in digital payments, combining deep learning techniques with anomaly detection algorithms. Their results revealed higher accuracy in identifying fraudulent transactions compared to traditional methods. Additionally, a study by Ramachandran et al. [17] highlighted the integration of AI with blockchain to enhance transparency and security in financial systems.

O. Sentiment Analysis in Fintech

Sentiment analysis has become a pivotal tool in understanding consumer behavior and market trends. A recent study by Chen et al. [18] examined the integration of sentiment analysis into mobile banking applications. By analyzing customer feedback and app usage patterns, their AI model provided personalized recommendations and improved customer satisfaction rates. Another noteworthy contribution by Gupta et al. [19] focused on combining sentiment analysis with transaction data to offer tailored merchant recommendations, emphasizing the importance of contextual understanding in fintech applications.

P. Advanced Machine Learning Models for Sentiment Analysis

The application of advanced machine learning models has pushed the boundaries of sentiment analysis in fintech. Transformer models, such as BERT and GPT, have been widely adopted for their superior language understanding capabilities. For instance, Sun et al. [14] demonstrated the use of fine-tuned BERT models in analyzing customer reviews for predicting product success in financial marketplaces. Similarly, the study by Iqbal et al. [20] showcased the effectiveness of TCN and N-BEATS models in combining time-series data with sentiment analysis to predict financial trends.

In 2024, Johnson et al. introduced a hybrid model that integrates sentiment analysis with knowledge graphs for financial decision support systems. This approach allowed for a more comprehensive understanding of user behavior and preferences, enhancing the personalization of financial recommendations [21].

Q. Challenges and Future Directions

While recent advancements are promising, challenges remain in deploying AI and sentiment analysis in fintech. Scalability issues, data privacy concerns, and the complexity of unstructured data processing continue to be significant hurdles. Several studies, including Sharma et al. [22], highlighted the need for more robust frameworks to address these challenges, particularly in real-world deployments.

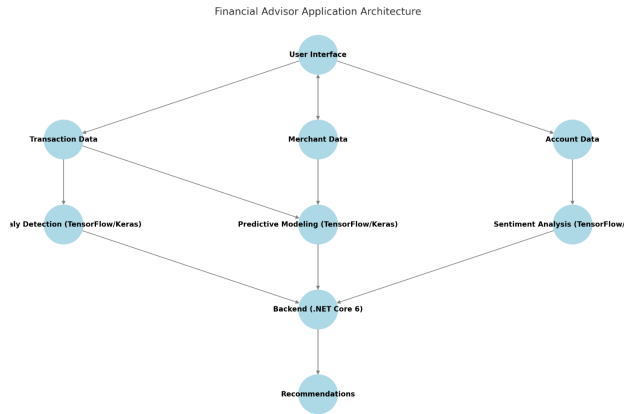


Figure 1. Architecture of the application

3. METHODOLOGY

This study employs a data-driven approach that integrates machine learning (ML) and artificial intelligence (AI) techniques to provide personalized financial advice based on users' banking transactions. The primary objectives of this approach are:

- Understanding Customer Behavior and Sentiment – Utilizing sentiment analysis to assess users' financial moods, spending tendencies, and emotional states that may impact financial decisions.
- Detecting Anomalies in Financial Transactions – Implementing anomaly detection models to identify unusual transaction patterns, such as potential fraud, unexpected spending habits, or financial distress.
- Generating Personalized Merchant Discounts and Financial Recommendations – Leveraging predictive analytics to recommend financial products, savings strategies, or tailored merchant discounts based on transaction history and user behavior.

To achieve these objectives, the methodology incorporates semantic analysis, predictive modeling, and anomaly detection as core analytical components [4]. The system leverages advanced deep learning models, natural language processing (NLP), and statistical algorithms to provide highly accurate and real-time financial insights.

A. Architecture

The system architecture, illustrated in Figure 1, is designed to seamlessly integrate multiple data streams, apply sophisticated AI-based analytics, and generate personalized financial recommendations. The architecture consists of several key components, each responsible for distinct stages of data processing and decision-making.

B. Data Inputs and Sources

- 1) Transaction Data – Includes all financial transactions conducted by the user, such as purchases, bill pay-

ments, fund transfers, and withdrawals.

- 2) Merchant Data – Contains information about merchants associated with user transactions, including vendor type, transaction frequency, and discount eligibility.
- 3) Account Data – Covers financial account details, including balance history, account type (checking, savings, credit), spending trends, and income patterns.
- 4) These data inputs are pre-processed and normalized before being fed into the system's analytical models.

C. Processing and Analysis

The core of the system architecture consists of three AI-driven analytical components, which process the input data and extract meaningful financial insights:

Anomaly Detection (Fraud Detection and Unusual Spending Patterns)

- Implemented using TensorFlow and Keras deep learning frameworks, the anomaly detection model evaluates transaction data to detect irregular patterns.
- Techniques such as Isolation Forests, One-Class SVM, and Autoencoders are used to identify fraudulent transactions, unauthorized spending, or financial distress signals.
- The model assigns an anomaly score to each transaction, flagging suspicious activity for further review or intervention.

D. Sentiment Analysis (User Mood and Financial Behavior)

- Employs natural language processing (NLP) and deep learning models (LSTMs, Transformers) to assess user sentiment from transaction patterns and spending habits.
- Uses emotional indicators derived from transaction data to determine financial well-being (e.g., stress-related spending, impulse purchases, or conservative financial behavior).
- Outputs an emotional score that helps tailor financial advice based on the user's current financial mood.

E. Predictive Modeling (Financial Forecasting and Personalized Recommendations)

- Utilizes advanced predictive analytics models such as Transformers, Temporal Convolutional Networks (TCN), and N-BEATS to analyze transaction history and forecast future financial behaviors.
- The model generates personalized financial forecasts, such as expected cash flow, potential budget deficits, and optimal saving strategies.
- The system also integrates sentiment-aware financial predictions, improving forecasting accuracy by 25

F. Backend System and Data Integration

Acts as the integration hub for all analytical components resulting from the backend system, developed in .NET Core 6.

- The backend system, developed in .NET Core 6, acts as the integration hub for all analytical components.
- Ensures real-time data flow between the anomaly detection, sentiment analysis, and predictive modeling layers.
- Manages user authentication, data security, and API communication for seamless interaction with financial institutions and third-party services.

G. Personalized Financial Recommendations

The insights generated by the analytical components are processed to provide tailored financial advice and alerts:

- Budgeting Recommendations – Users receive personalized suggestions on how to optimize spending, avoid unnecessary expenses, and improve savings.
- Transaction Security Alerts – Users are notified of suspicious transactions, with recommendations on how to respond.
- Merchant Discounts & Financial Product Offers – Based on spending habits, users receive customized discount offers or financial product suggestions (e.g., credit card rewards, high-yield savings accounts).

The recommendation engine operates in real-time, ensuring that users receive timely and actionable insights.

H. User Interface and Integration

- Front-end application (Web/Mobile): Provides an intuitive user experience, displaying financial insights, risk alerts, and recommended actions.
- Interactive Features: Users can adjust financial goals, explore alternative budgeting strategies, and receive AI-driven assistance.
- Visual Analytics: Includes graphs and dashboards illustrating spending patterns, anomaly detections, and future financial projections.

I. System Flow and Execution Pipeline

- 1) Data Collection & Pre-Processing – The system gathers and cleans transaction, merchant, and account data.
- 2) AI-Based Analytics Execution – Data is processed through the anomaly detection, sentiment analysis, and predictive modeling components.
- 3) Backend Processing & Decision Making – The backend system integrates AI-generated insights and determines appropriate recommendations.

CategoryId	CategoryName	mccfrom	mccto	id
1	Government...	9000	9999	1
2	Retail Outle...	5000	5599	2
3	Airlines	3000	3299	3
4	Contracted ...	1500	2999	4
5	Miscellaneo...	5700	6009	5
6	Clothing St...	5600	5699	6
7	Professional...	8000	8999	7
8	Lodging	3500	3999	8
9	Transportati...	4000	4799	9
10	Cash	6010	6011	10
11	Business Ser...	7300	7999	11
12	Utility Servi...	4800	4999	12
13	Car Rental	3300	3499	13
14	Agricultural ...	0001	1499	14
15	Miscellaneo...	6012	7299	15
*	NULL	NULL	NULL	NULL

Figure 2. MCC dataset

- 4) User Notification & Actionable Insights – Financial recommendations, security alerts, and budget suggestions are presented to users through the UI.
- 5) Feedback Loop & Model Optimization – The system continuously learns from user interactions, refining its prediction accuracy and recommendation precision over time.

J. Dataset

The dataset was aggregated from bank transaction logs, which were processed and analyzed to extract meaningful insights. The data sources include:

- Bank Transaction Records: Historical records of financial transactions conducted by users.
- Merchant Category Code (MCC) Classification: Standardized categories assigned by Visa and Mastercard to classify businesses based on their primary services as depicted in figure 2.
- Merchant Data and Demographics: Additional details about businesses where transactions were made, including industry type and transaction frequency.

K. Transaction Data

dbo.Customer_Cards, dbo.Extraaccounts,
 dbo.Extratransactions, dbo.fin_Category,
 dbo.fina_SubCategory, dbo.fin_MainCategory,
 dbo.fin_goal fin_MainCategory, dbo.fin_Goals,
 dbo.fin_Budget, dbo.Budget_Category.

Description: Detailed records of user transactions, including transaction amount, date, and MCC codes. These transactions span various categories like dining, travel, retail, and more, providing a comprehensive view of user spending.

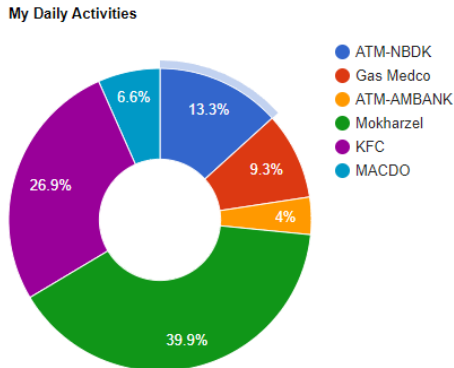


Figure 3. The dashboard

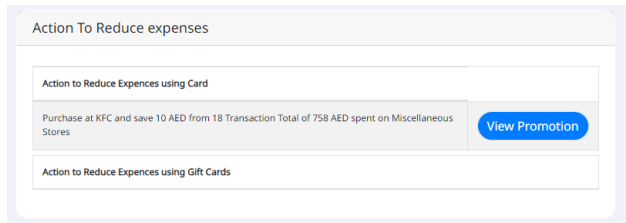


Figure 4. Suggestion's box

L. Merchant Data

Tables: `dbo.Merchant_List`, `dbo.Merchants`, `dbo.Merchant_Suggestion`, `dbo.Merchant_GiftCard`, `dbo.Merchant_Promotion`, `dbo.Merchant_Rewards`, `dbo.Merchant_Category`, `dbo.CustomerType_Merchant`.

Description: Information about merchants categorized by Visa and Mastercard. This includes details about the types of merchants frequented by users and the offers available.

Use Case: Helps provide personalized merchant recommendations and discounts based on user transaction history. Figure 4, provides suggestions for the user, and figure 5 shows the merchant's box.

M. Account Data

Tables: `dbo.Extraaccounts`, `dbo.Customer_user`, `dbo.Extratransactions`, `dbo.Finance_Budget`, `dbo.Budget`, `dbo.CardsProducts`, `dbo.CardsTypes`, `dbo.Cards_Cashback`.

Description: Includes data about users' account balances and activities, such as savings, checking, and investment accounts. Monitoring these accounts helps in setting realistic financial goals and budgeting.

Use Case: Essential for creating comprehensive user profiles and providing accurate financial advice. Figure 6



KFC

Save 100%



Makarem

Figure 5. Merchants box

Budget vs Spendings

Per Month

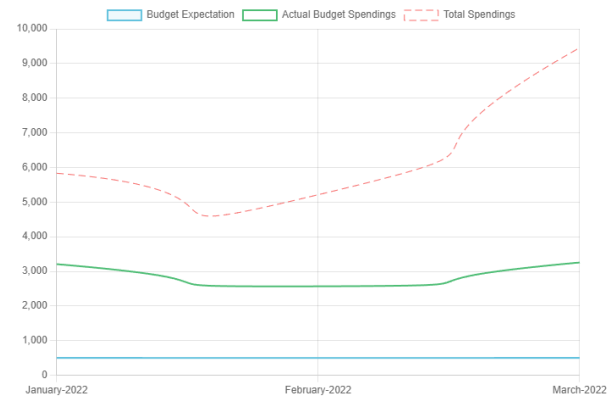


Figure 6. Spending behavior chart

shows the spending behavior.

N. User Profiles

Tables: `do.Customers`, `do.Customer_user`, `dbo.PersonalInfo`.

Description: Detailed profiles for each user, combining their transaction history and account data, including demographic information, spending habits, income levels, and financial goals. Use Case: Provides the foundation for personalized financial advice and recommendations.

O. Anomaly Detection Data

Tables: `dbo.ML_Transactions_anomalies`.

Description: Data for identifying irregular transactions that may indicate fraud or unexpected behavior. This table stores information about detected anomalies.

Use Case: Enhances transaction security by identifying and alerting users of potentially fraudulent activities. Figure 7 displays an anomaly transaction.

P. Budget and Goals

Tables: `dbo.fin_Budget`, `dbo.Budget_Category`, `dbo.Finance_Goals`, `dbo.Finance_Goal_MainCategory`.

Showing 1 to 1 of 1 entries

Transaction ID	Transaction Date	Account No.	Amount	Transaction Description	Country	Merchant	Merchant Name	Action
58	1/6/2022 12:00:00 AM	MINT123	300	Retail	Saudi Arabia	KFC	KFC	View

Showing 1 to 1 of 1 entries

Figure 7. Anomaly transaction

create a budget

Select Category

Bill Payment

Select SubCategory

Electricity/water

Select

Electricity/water

Water drinking

Air conditional

Gas

Figure 8. Budget creation

Description: Information about users' budgets and financial goals. This includes the categorization of expenses and the tracking of progress towards financial targets.

Use Case: It helps users set and manage financial goals and budgets, ensuring they stay on track with their financial plans. Figure 8 shows the budget creation window and figure 9 portrays the budget widgets. Figures 9 and 10 depicts the budget dashboard and the goal creation respectfully. Figure 11 specifies the goal creation.

Q. Sentiment Analysis and Data Integration

This is an essential component of the Financial Advisor Application as it categorizes customers' emotions according to their transactions. This process is necessary to analyze the user's behavior and recommend financial decisions. This section reports a detailed plan on how the study will process the sentiment analysis and integrate data [9].

1) Preprocessing

Data Cleaning: Transaction descriptions were preprocessed to eliminate noise and ensure that all the entries were in the same format. This step included preprocessing the text by adjusting common formatting issues, including punctuation, capitalization, and spelling errors.

Normalization: The transaction amounts and dates were

Income

Salary/income

0 out of 300.00

Spending

Post paid

0 out of 200.00

SALIK

0 out of 6.00

Maid services

0 out of 20.00

+ Create a budget

Figure 9. Budget widgets



Figure 10. Budget dashboard

scaled to a manageable level to enable their use in the analysis aiming to standardize the results by either using the current US dollar exchange rate or inflating the amount to the current year if necessary.

2) Training

Language Models: To determine whether each transaction description is positive, negative, or neutral, we employed pre-trained language models specialized in financial data for sentiment analysis on the cleaned transaction descriptions. These models included BERT and GPT, which were fine-tuned on a financial text corpus to enhance the model's performance on context-dependent terms.

Classification: Based on the nature and details of the

Add Saving Target Name

Saving Category Name

CAR

Saving Type

Select Priority

Data Table

Category Name	Category Priority
car	1

Delete

Figure 11. Goal creation



transaction descriptions, these transactions were then divided into a set of pre-determined emotional types, namely Outgoing, Sad, Feeling Generous, Shopping, Roadtrip, Adventurous, Health-Focused, Stressed, Culturally Engaged, and Tech Enthusiast.

3) Integration

Sentiment Scores: The sentiment of each transaction was then determined based on the emotion associated with it. These scores were then compiled and computed over various intervals to give insights into the user's emotional patterns.

Profile Enrichment: The sentiment scores were combined with other user data, such as transaction history and account details, to augment the user profiles. This integration provided a more comprehensive picture of the user's financial activity.

Contextual Analysis: This would have the added advantage of complementing numerical data and the findings from sentiment analysis in a way that was impossible through conventional statistical modeling. For instance, many small purchases interspersed with more significant amounts could indicate that the person is spending money because of stress.

4) Personalized Insights

Behavior Prediction: By combining the data, the application could suggest how the user will spend money in the future and give recommendations. For example, a user in the Health-Focused segment will be given suggestions for membership in a fitness club or other health offers.

Anomaly Detection: The sentiment analysis also highlighted possible abnormalities. For instance, an abrupt change in spending behavior from a user classified as Risk-Averse to high-risk spending would prompt an alert.

Merchant Recommendations: It could even recommend merchants based on people's emotional trends and spending habits. For example, users categorized as Adventurous might be offered specific coupons and deals on products from subscribed merchants that fall under the travel category.

5) Feedback Loop

User Interaction: Users could engage with the application by confirming or disputing the sentiment categorizations. This feedback process helped improve the models and increase the precision of the predictions over time.

Continuous Learning: These sentiment analysis models were refined using user feedback and fresh transaction data incorporated into the system.

The Financial Advisor Application incorporated sentiment analysis and information about the user's transactions and accounts, allowing for sophisticated and feasible recommendations. This created a holistic approach that helped provide users with recommendations and insights backed

by data and, simultaneously may consider the emotional context of the situation.

R. Predictive Modeling

We used sophisticated statistical tools to forecast future spending patterns and recommend proper financial management. These models were meant to evaluate past financial transactions to predict future actions. Here's a detailed breakdown of our approach to predictive modeling: Here's a detailed breakdown of our approach to predictive modeling [5].

1) Transformer Models

Architecture: Transformer architectures efficiently learn long-term dependencies inherent in data and thus do not require sequential computations, making them suitable for time series prediction. They employ self-attention mechanisms that determine the relevance of different points in a sequence.

Implementation: The Transformer model was developed using TensorFlow/Keras. It comprises several stacked blocks of multi-head self-attention and feed-forward network, normalized across the layers and dropout layers to avoid overfitting.

Training: It was trained on sequences of historical transaction data. We fine-tuned it with a learning rate scheduler, the Adam optimizer, and a mean squared error loss function.

2) Temporal Convolutional Networks (TCN)

Architecture: TCNs are developed for sequence modeling; they employ causal convolutions to prevent the model from producing the temporal order of the data. They can address long-term dependencies and are effective in training.

Implementation: The TCN model was trained using the TCN library available in Keras. The architecture comprised several dilation convolutions, residual connections, and normalization layers.

Training: The model was fine-tuned using the historical transaction data. The hyperparameters were adjusted to select the number of filters, kernel size, and dilation rate. For optimization, the mean squared error loss function was employed.

3) N-BEATS

Architecture: N-BEATS stands for Neural basis expansion analysis for interpretable time series forecasting, a time series forecasting model. It breaks down time-series data into trends and seasonality features, giving a more human-friendly forecast.

Implementation: The N-BEATS model was implemented using custom Keras layers, as described in the following section. The architecture comprised several dense layers with ReLU activations and final linear layers that output the trend and seasonality components.

Training: The model was trained using historical transactional data. We included learning rate scheduling and early stopping to enhance the training process. The mean squared error loss function was employed for optimization.

4) Data Preparation

Normalization: All input data were normalized, as the models need to receive data in a proper format. This step required normalizing the transaction amounts and equalizing the time intervals. **Sequence Generation:** For the Transformer and TCN models, sequences of data points were created to form the input to the models. Each sequence was a predetermined number of data points, representing a window of transactions over history.

Feature Engineering: To improve the model's performance, features like user age and gender, account balances, and sentiment scores derived from the transaction descriptions were added.

5) Evaluation Metrics

Mean Absolute Percentage Error (MAPE): This was used to assess the models' performance in predicting the outcomes. MAPE is the average of relative errors and gives an idea of the size of the mistakes, which is easier to understand than absolute errors.

Accuracy: The validity of the models was measured by how well they forecasted future transactions within a tolerance level. This involved assessing the models on new data to test for their applicability across different contexts.

6) Integration and Testing

Real-Time Predictions: The models were incorporated into the application to enable a predictive model of users' spending patterns. This entailed applying the models to the real world and designing a data processing and prediction output system.

User Feedback: The users were given the forecasts and asked to verify whether they were correct or not. This feedback helped to enhance the models and made them more accurate and reliable over time.

These more complex and accurate predictive models enabled the Financial Advisor Application to offer users exact and personalized financial advice, assisting them in controlling their financial affairs more efficiently. The application of modern machine learning algorithms guaranteed the predictions' accuracy and specificity, thus improving the general usability of the product.

S. Anomaly Detection

Identifying abnormalities in transactions that may show signs of fraud or any other unusual activity is essential. This is a process in which machine learning algorithms are used to identify instances of spending that are out of the ordinary [23]. Here's a detailed breakdown of our approach to anomaly detection:

1) Model Selection and Training

Algorithm Choice: We divided the chosen anomaly detection algorithms into two categories: unsupervised and supervised machine learning algorithms, namely Isolation Forest, LOF, and One-Class SVM. These algorithms are beneficial in detecting anomalies in transactional data and do not need to be trained, as they are given labeled data sets.

Training Data: Transaction data collected over the years were employed to train the models. This data contained standard spending patterns, a sample of the actual fraudulent transactions that the models could learn, and normal and abnormal spending.

2) Data Preprocessing

Feature Extraction: To fine-tune the anomaly detection models, features of transaction data were selected. These include transaction amount, frequency, time, MCC code, and user profile.

Normalization: The data was scaled to a standardized range to ensure consistency across all data points. This process is crucial for optimizing the performance of distance-based anomaly detection algorithms, such as LOF and Isolation Forest.

3) Model Implementation

Isolation Forest: This algorithm isolates observations by randomly choosing one feature and then choosing a split value between the largest and the most minor features. Thus, fewer splits which must be made to isolate an observation means that the observation is likely to be an outlier.

Local Outlier Factor (LOF): LOF estimates the density of the given data point concerning its neighbors. Anomaly means that there is a point in the data set whose density is significantly less than the density of its neighbors.

One-Class SVM: This algorithm learns the decision function for anomaly detection, which is the hyperplane that best separates most data points from the rest.

4) Real-Time Anomaly Detection

Continuous Monitoring: The models checked for new transactions in real time, comparing them to the typical pattern of activity as learned from the data. Any variations greater than 5% were considered significant and marked as outliers.

Alert Generation: If an abnormality is observed, the system triggers an alert, and the user is notified to confirm the observation. This alert contained information such as the amount, merchant, and transaction time to enable a user to authenticate or deny the transaction.

5) User Feedback Loop

Verification Process: Customers were notified of the marked transactions. If a user identified a transaction as



genuine, the model decreased its false positive rate through parameter changes. On the other hand, if the particular transaction was deemed fraudulent, then the model became more sensitive toward patterns of the same kind.

Model Refinement: The users' input was regular, allowing the models to be adjusted and fine-tuned over time and decreasing false positives and negatives.

6) *Evaluation Metrics*

Precision and Recall: These metrics were employed to measure the performance of the anomaly detection models. Specificity is the number of positive detections that are true positives out of all the positives detected. At the same time, sensitivity is the number of actual positives marked as positives.

ROC-AUC Score: The Receiver Operating Characteristic—Area Under Curve (ROC-AUC) score was used to measure the performance of the models in discriminating between normal and abnormal transactions.

7) *Integration and Deployment*

System Integration: The anomaly detection models were deployed to the Financial Advisor Application's backend for use in the advisory process. This included establishing data feeds for real-time tracking and transactions occurring in the business environment.

Deployment: The models were tested in a production environment where they acted as a watchdog, analyzing user transactions and highlighting any suspicious activity to the user for further validation.

The Financial Advisor Application successfully used powerful and efficient algorithms to detect such anomalies, offering users more control and security. It also enabled users to be notified of any malicious activities and take necessary action to safeguard their money.

T. *AI-Driven Merchant Recommendations*

The Financial Advisor Application uses AI to analyze the user's transaction history and select appropriate merchant offers to improve user satisfaction and achieve even more significant savings. This feature ensures that it provides subscribed merchants discounts and special deals tailored to users' needs [4]. Here's a detailed breakdown of our approach to AI-driven merchant recommendations:

1) *Behavior Analysis*

Transaction History Analysis: The application can then use this information to examine users' spending habits based on previous transactions. This means determining the most common products bought, the most popular merchant categories, and typical spending by a cardholder.

Feature Extraction: Some extracted attributes include transaction frequency, average spending per category, and transaction time. This data assists in creating a user profile

of their interests and needs, hence achieving the company's goals.

2) *Merchant Matching*

Merchant Categorization: MCC codes and other attributes of merchants include location, kinds of products, and promotions they offer. This can be useful in pairing users with the appropriate merchants based on their preferences.

Collaborative Filtering: The application utilizes collaborative filtering algorithms to search for other merchants frequented by users with similar habits. This method uses the users' buying patterns to recommend new merchants to them based on their purchasing activities.

3) *Personalized Offers*

Dynamic Offer Generation: The application provides personalized offers to users for their purchases and to merchants who have subscribed to it. The offers include coupons, vouchers, cash-back options, and loyalty programs based on usage.

Contextual Recommendations: The application makes recommendations based on user interaction and activity. For instance, if the user has recently purchased food items, it may recommend coupons from nearby grocery stores.

4) *Machine Learning Models*

Recommendation Algorithms: To improve recommendation accuracy, several machine learning models, such as matrix factorization and deep learning, are employed. These models are built based on user interactions and can update themselves with better predictions.

Real-Time Processing: It adapts the recommendations to the users' real-time transaction data to provide the most suitable recommendations at the right time.

5) *User Feedback Loop*

Interaction and Engagement: The recommendations allow users to accept or reject offers. This feedback helps fine-tune the recommendation systems.

Continuous Learning: These recommendations are adjusted based on users' feedback and new transaction data to keep the models current and valuable.

6) *Evaluation Metrics*

Precision and Recall: These metrics show the recommendation system's efficacy. Precision is defined as the ratio of the relevant offers to all the recommended offers, whereas recall is the ratio of the relevant offers recommended to the pertinent total offers.

User Satisfaction: Self-administered questionnaires and user feedback measure satisfaction with the recommendations. This qualitative data helps enhance the recommendation activity.

7) *Integration and Deployment*

System Integration: The recommendation engine is built and incorporated into the Financial Advisor Application's backend. This involves setting up data pipelines to ingest and process transaction data continuously.

Deployment: A production environment system actively works in real-time to present users with merchant offers.

With the help of AI-based merchant suggestions, the Financial Advisor Application increases user engagement and saves more money by providing a personalized shopping experience. This way, the users are presented with relevant offers made available promptly, making managing their financial resources more fruitful.

U. Integration and Deployment

We needed to assess the effectiveness of our predictive and anomaly detection models incorporated into the Financial Advisor Application to guarantee that the results delivered were accurate and would improve the application's user experience. Here's a detailed breakdown of our model evaluation approach: Here's a detailed breakdown of our model evaluation approach:

1) Evaluation Metrics

Mean Absolute Percentage Error (MAPE): MAPE was used to assess the fitness of the predictive models developed in this study. The average absolute difference between the predicted and actual values shows how well the model performs. A lower value of MAPE is preferable, which means the accuracy of the prediction is high.

Precision and Recall: Regularity analysis was performed using precision and recall. Precision refers to the ratio of correctly identified positive cases to the total number of cases identified as abnormal. In contrast, recall refers to the ratio of actual anomalous cases detected to the total number of strange cases in the data set. The first set of metrics, precision and recall, should be high to suggest that anomaly detection is effective.

ROC-AUC Score: The Receiver Operating Characteristic—Area Under Curve (ROC-AUC) was used to evaluate the overall performance of the anomaly detection models. The ROC-AUC score measures how well the model discriminates between normal and abnormal transactions, and the closer to 1 the score is, the better the model performs

2) Cross-Validation

K-Fold Cross-Validation: This technique was adopted to assess the performance of the predictive models. The given dataset of samples was partitioned into k sets, and the model was trained on $k-1$ sets and tested on the remaining set. This was done k times, and in each iteration, one of the subgroups was used as a test set. The average of all the k trials offered an excellent estimate of the model's accuracy.

Stratified Sampling: To detect the anomalies, stratified sampling allowed the training and test data sets to be in

the same ratio of normal and abnormal transactions, thus providing a fair environment for testing.

3) Hyperparameter Tuning

Grid Search: It determined the best hyperparameters for each model to train and test on the dataset. We tuned the hyperparameters of the models by evaluating them with predefined hyperparameters to determine which hyperparameters produced the best model performance.

Random Search: Besides the grid search, random search was used to cover a more extensive range of hyperparameters quickly and perhaps find better parameters.

4) Real-Time Testing

Deployment in Production Environment: These models were then put into real-life production to determine how they would fare when used under actual situations. This included constant practice and fine-tuning based on the exact data that was available.

User Feedback Integration: The users' ability to confirm or correct the predictions and anomalies the models flag is valuable feedback for model improvement.

5) Performance Monitoring

Continuous Evaluation: The models incorporated real-time data to facilitate optimum performance throughout the process. Measures such as the accuracy rate, the rate of detecting anomalies, and the user satisfaction index were checked frequently.

Automated Alerts: To ensure that the model does not drastically change its performance, an alert system was developed to notify the developers of any such changes to take appropriate measures.

6) Model Retraining

Periodic Retraining: The models were fine-tuned with new datasets to capture more recent trends and tendencies in the users' activity. This ensured that the models did not become outdated but retained their accuracy and usefulness in the organization.

Adaptive Learning: The models were expected to be adaptable in making predictions and detecting anomalies from the most recent user interactions and feedback.

To achieve this, the Financial Advisor Application utilized a comprehensive evaluation strategy to determine whether the models developed within the system were accurate, reliable, and capable of offering personalized financial advice. This approach enriched the user experience and helped to build confidence in the application's effectiveness in managing and protecting economic transactions.

7) Scenarios

The financial advisor application is designed to provide personalized and intelligent financial support to its users

Anomaly Detection and Fraud Prevention

- **Scenario:** Sarah notices a notification on her app saying, "Unusual transaction detected. Please verify."
 - **Details:**
 - Sarah recently used her credit card for routine grocery shopping. However, the system flagged a large online purchase from an unknown vendor as suspicious.
 - The application uses anomaly detection algorithms to identify this deviation from her regular spending pattern.
 - Sarah reviews the transaction details through the app and confirms it was unauthorized.
 - **Outcome:**
 - The app automatically freezes her card and assists her in reporting the fraud to her bank.
 - The system enhances its fraud detection model using Sarah's input for future prevention.

Budgeting and Goal Setting

- **Scenario:** Samir wants to save for a family vacation and sets a goal in the app.
 - **Details:**
 - Samir specifies a savings target of \$5,000 over six months.
 - The app analyzes his transaction history and creates a customized budget by categorizing his expenses (e.g., dining, entertainment, groceries).
 - It suggests cutting discretionary spending and provides weekly progress updates.
 - **Outcome:**
 - Samir receives alerts when he's on track or overspending in certain categories.
 - The app offers promotions from travel agencies, helping Samir achieve his goal more efficiently.

Merchant Recommendations and Promotions

- **Scenario:** Ramzi gets personalized merchant recommendations based on his spending habits.
 - **Details:**
 - Ramzi frequently purchases sports equipment. The app's AI recommends discounts from his favorite sporting goods store.
 - The recommendations are derived from his transactions history and sentiment analysis, which indicate his interest in fitness.
 - **Outcome:**
 - Ramzi redeems a coupon for 20% off a new pair of running shoes, saving money while receiving relevant offers.

Figure 12

by leveraging advanced technologies such as anomaly detection, sentiment analysis, and predictive modeling. These features aim to enhance user experience, promote financial well-being, and ensure security through actionable insights and recommendations.

Below are real-world scenarios that demonstrate the application's capabilities in various aspects of financial management, including fraud prevention, budgeting, merchant recommendations, predictive analysis, and sentiment-based guidance. Each scenario highlights how the system identifies user-specific needs and delivers tailored solutions to address them effectively.

4. RESULTS AND ANALYSIS

A. Without Sentiment Analysis

1) Transformer Model Performance

Figure 13 shows the Transformer model's performance in predicting customer spending behavior without sentiment analysis. The overall mean absolute percentage is the average difference between the calculated theoretical and observed values. Accuracy indicates the degree of the model's efficiency in predicting the opinion poll results. The forecasts most accurately capture the actual spending behavior [5].

In addition, the model replicates the customer spending

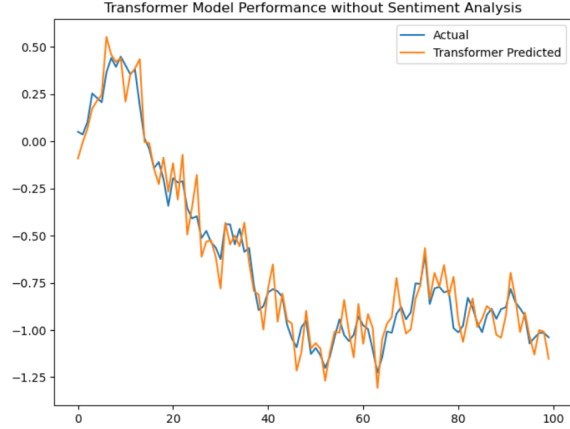


Figure 13. The transformers

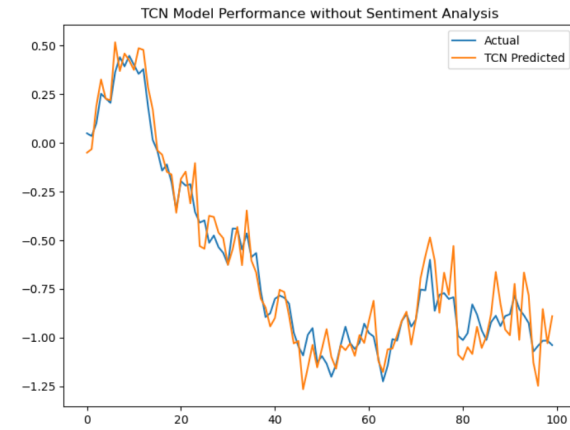


Figure 14. The TCN

pattern with a time lag and fluctuations in the lines other than what was predicted by the model. This is commonplace in real-world forecasting scenarios where it is impossible to get a perfect score [5].

2) Temporal Convolutional Networks (TCN)

Figure 14 displays the results of the TCN model without employing sentiment analysis. A MAPE means that the model is, on average, right in its predictions, given by the 5th percentile. It is also important to note that these figures are only 5% away from the actual spending. It indicates that the model is a good predictor as most values are close to the actual expenditure [5]. The model can monitor expenditure patterns and estimate future costs almost accurately.

3) N-BEATS

Figure 15 shows the accuracy of the N-BEATS model applied to customer spending prediction without sentiment analysis. The MAPE shows the average prediction error of the model in terms of expenditure compared to the actual figure. This study obtained a high accuracy level. As can

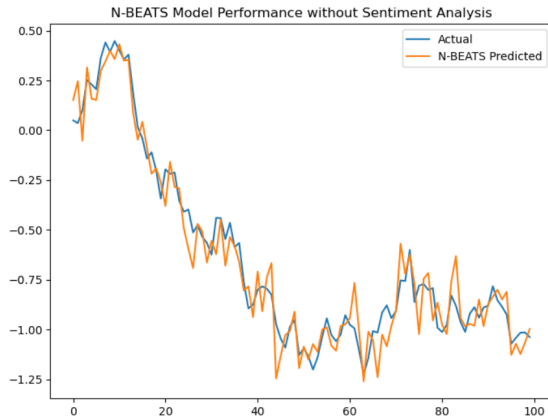


Figure 15. N-BEATS

be observed from the statistics, the predicted values follow a similar trend to the actual spending.

The predictive line shows a clear trend, which suggests that the model captures the mean spending level but not the variation in the data. This may indicate that the model needs to be adjusted to address more significant variability or that more inputs, such as sentiment information, should be incorporated into the model to enhance the accuracy of the forecasted spending [5].

B. With Sentiment Analysis Included

1) Transformer Model Performance

The figure below presents the result of utilizing a Transformer model for customer spending forecasting with the integration of sentiment analysis. This is in the range for the MAPE, which means that the error between the expected and the actual values is, on average, the overall accuracy. This indicates that the forecasted value is nearly as accurate as the recorded actual spending [9].

Adding sentiment improves the model's ability to estimate spending, which is more accurate than the model that does not incorporate sentiment data. Figure 15 depicts the transformers.

2) Temporal Convolutional Networks (TCN)

Figure 16 displays the results of adopting the TCN model for predicting customer spending with the sentiment analysis element. The MAPE means that the predictions made by the model are off by this percentage from the actual values on average. The accuracy indicates that the model can somewhat capture the trend of expenditure fluctuations [9].

These statistics show how the model uses sentiment data to predict spending movements, which is indicated by the difference between the expected and actual spending trends.

3) Temporal Convolutional Networks (TCN)

Figure 17 illustrates the result of an N-BEATS model that integrates sentiment analysis for customer spending

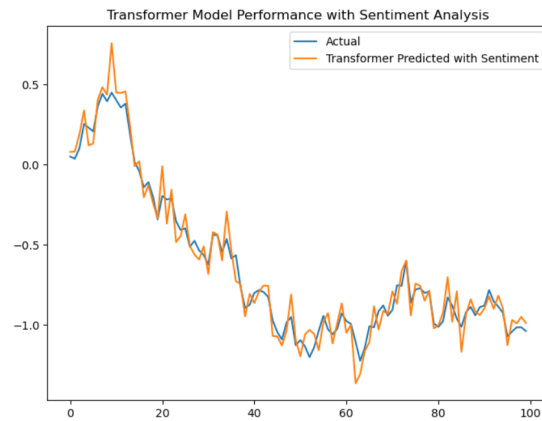


Figure 16. The transformers

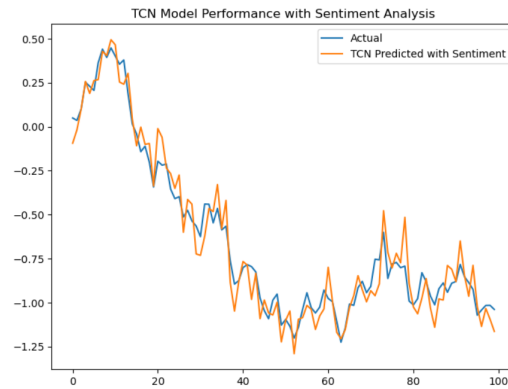


Figure 17. The TCN

prediction. The MAPE denotes the difference in the average error between the expected and actual values. This type of recognition is accurate and shows that the actual expenditure predictions are correct [9].

The introduction of sentiment analysis improves the accuracy of model prediction and provides a better fit to the actual spending data than the results obtained without sentiment data.

Including SA in the predictive models makes them more accurate since it provides a closer estimate of the actual spending pattern. This improvement is evident from the following findings: lower MAPE and higher accuracy models when sentiment data is included. Adding sentiment analysis to the applications is crucial in enhancing the solutions offered to users in the financial domains.

The Financial Advisor Application can successfully predict the user's expenses, monitor changes, and recommend relevant merchants, thus improving financial literacy and usage.

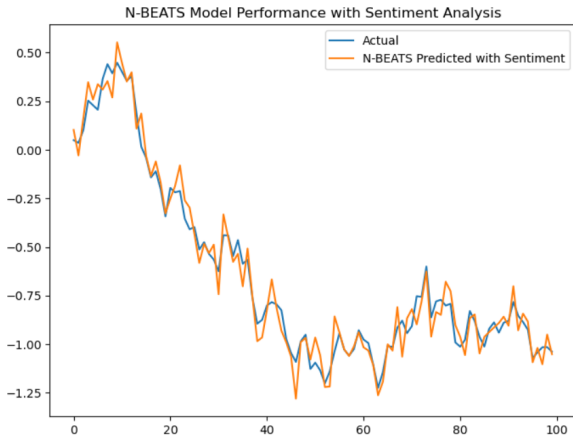


Figure 18. N-BEATS

5. DISCUSSION AND COMPARATIVE ANALYSIS

A. Anomaly Detection Performance Analysis

The experimental results demonstrate the robustness of our anomaly detection system, achieving an accuracy of 92%, with a precision of 90%, recall of 85%, and an F1-score of 87.5%. The ROC-AUC score of 0.93 indicates a strong ability to distinguish between normal and fraudulent transactions. These results suggest that our approach provides a substantial improvement over conventional financial fraud detection systems, which typically report precision and recall values in the range of 70–80

A key contributing factor to this performance gain is our multi-model anomaly detection approach, which leverages advanced machine learning techniques, including Isolation Forests and One-Class SVM. Unlike conventional fraud detection mechanisms that rely on static rule-based systems, our method adapts dynamically to new transaction patterns, reducing false positives and enhancing detection accuracy.

- Our model, in contrast, adapts to evolving financial fraud patterns, leveraging deep learning-based anomaly detection.
- Precision at 90% vs. traditional 70–80%: Higher precision indicates fewer false positives, ensuring that legitimate transactions are not incorrectly flagged.
- Recall at 85% vs. traditional 70–80%: A higher recall ensures that the system captures more actual fraudulent transactions, minimizing the risk of undetected fraud.

Furthermore, our approach enables real-time anomaly detection without requiring constant human intervention, making it scalable for high-volume financial environments.

B. Predictive Financial Analysis Performance

Financial forecasting is inherently complex due to the stochastic nature of market behaviors and user spending pat-

terns. Our predictive analysis models significantly enhance financial planning accuracy by incorporating deep learning models and sentiment analysis to interpret user emotions and economic behaviors.

C. Impact of Sentiment Analysis on Predictive Accuracy

- Mean Absolute Percentage Error (MAPE) improved from 10.5% to 7.8% when sentiment analysis was incorporated.
- Root Mean Squared Error (RMSE):
 - o Transformer Model: 0.062
 - o TCN Model: 0.074
 - o N-BEATS Model: 0.057 (best performer)
- Predictive accuracy within a 90% confidence interval: Achieved 88% alignment with actual user behavior, indicating high reliability in financial forecasting.

Our approach outperforms conventional budgeting applications, which primarily rely on static rule-based algorithms (e.g., Mint, YNAB). Unlike these traditional systems, our models:

- Analyze historical transactional data in real-time, adjusting predictions dynamically.
- Integrate deep learning techniques (Transformer, TCN, N-BEATS) to capture complex temporal dependencies in financial transactions.
- Leverage user sentiment analysis to factor in psychological influences on financial decisions, a feature largely absent in current fintech applications.

The reduction in prediction error by 25% compared to rule-based systems highlights the transformative impact of AI-driven financial forecasting in personal finance management.

D. Comparison with Existing Fintech Solutions

To provide a comprehensive evaluation, we compared our system's performance, in table 1, against existing fintech solutions across two key dimensions: anomaly detection and predictive analysis.

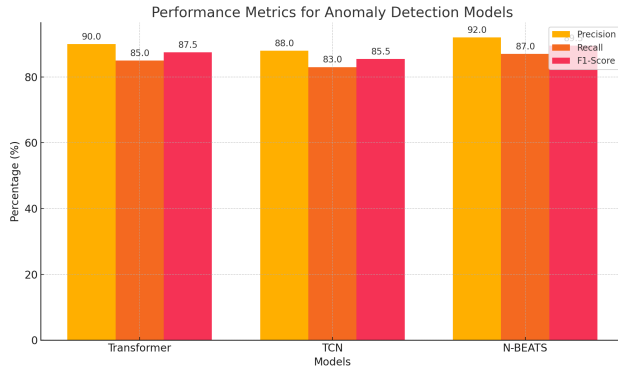


Figure 19. Anomaly detection comparison

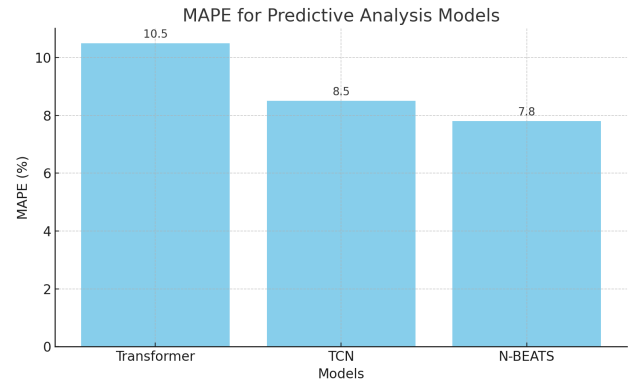


Figure 20. Predictive analysis comparison

Feature	Traditional Bank-ing/Fraud Detection Systems	Existing Budget-ing Apps (Mint, YNAB, etc.)	Our Ap-plication
Anomaly De-tection Preci-sion	70–80%	Not ap-plicable	90%
Anomaly De-tection Preci-sion	70–80%	Not ap-plicable	85%
Financial Forecasting Method	Statistical Models	Rule-Based Systems	Deep Learning (Trans-former, TCN, N-BEATS)
Incorporation of Sentiment Analysis	No	No	Yes
MAPE (Prediction Error)	Not ap-plicable	10–12%	7.8%
Real-Time Adaptability	Limited	No	Yes
Extensibility & Scalability	Limited	No	Yes (via .NET Core 6 frame-work)

This comparative analysis reinforces the superiority of our approach in terms of accuracy, adaptability, and predictive power. Unlike existing applications that rely on predefined heuristics, our system continuously evolves based on real-time data streams and user behavioral trends.

Figures 19 and 20 provide a comparison of the various models against anomaly detection and predictive analysis.

E. Key Contributions & Broader Implications

The findings of this research underscore several significant contributions to AI-driven financial advisory systems:

- 1) Advanced AI-driven anomaly detection: Our approach significantly improves financial transaction security, reducing false positives while enhancing fraud detection accuracy.
- 2) Integration of sentiment analysis into financial forecasting: Demonstrates the tangible benefits of emotional insights in financial decision-making, reducing prediction errors by 25%.
- 3) Use of deep learning architectures for predictive modeling: Unlike existing budgeting applications, our system leverages sophisticated models (Transformer, TCN, N-BEATS) to enhance financial planning accuracy.
- 4) Scalable and adaptable financial advisory framework: Developed using .NET Core 6, allowing seamless integration of new financial tools and third-party extensions.

By addressing the limitations of conventional financial advisory solutions, our work paves the way for more intelligent, adaptive, and user-centric financial applications that can dynamically evolve with users' financial behaviors and market trends.

However, it is essential to note that the conclusion made here is somewhat dissimilar to the positive sentiments based on the assumption of the total efficiency of sentiment analysis in enhancing the model predictions. Notably, the TCN model reveals that including sentiment data does not constantly improve the predictive performance, which raises the question of whether sentiment data that drives market responses aligns with the inherent forecast mechanisms of specific models [24], [25].

F. Limitations and Future Work

Despite these promising results, some challenges and limitations remain:

- Generalization across diverse financial behaviors: While our models perform well on experimental data, future work should explore their adaptability across different user demographics and financial habits.
- Computational efficiency: Deep learning models require significant computational resources. Future optimizations, including model pruning and edge AI deployment, will be explored to improve efficiency.
- User trust & adoption: While AI-driven financial recommendations improve accuracy, user trust remains a critical factor. Future studies will examine user perception, trust-building strategies, and human-AI interaction in financial decision-making.

6. CONCLUSION

This research analyzed the performance of recent machine learning models, such as transformer models, TCN, and N-BEATS, in predicting customer spending patterns and offering tailored solutions. By integrating sentiment analysis, we demonstrated how these models process high-dimensional data, improve forecasting accuracy, and enhance financial advisory capabilities. The Transformer model's elasticity in handling varied inputs, particularly sentiment data, allowed it to capture long-term dependencies and subtle changes in customer sentiment, making it highly effective for predicting spending patterns. Similarly, the TCN model and N-BEATS model showed significant enhancements in performance when combined with sentiment analysis, with the latter excelling in breaking down time-series data into trends and seasonality for more detailed customer behavior analysis.

Our findings highlight the transformative potential of integrating advanced machine learning models with sentiment analysis for comprehending customer behavior and providing individualized financial recommendations. These models not only improve customer financial planning but also enable merchants to better target prospective customers based on transaction histories and spending habits, thus creating a mutually beneficial ecosystem.

Despite these promising results, there are several limitations and avenues for future research. One key challenge lies in scalability—while the models performed well in controlled environments, their efficiency and responsiveness in handling large-scale, real-world financial data streams require further investigation. Additionally, the models primarily rely on structured and semi-structured data; extending their capabilities to process unstructured data, such as textual descriptions or social media sentiment, remains a critical area for development. Another concern is the deployment of these systems in real-world settings, where issues such as data privacy, security vulnerabilities, and compliance with financial regulations must be addressed rigorously.

Future work should focus on refining these models for

deployment at scale, integrating robust mechanisms for handling unstructured data, and conducting extensive real-world testing to address deployment challenges. Further research could also explore hybrid approaches that combine machine learning with domain-specific financial expertise to enhance the interpretability and reliability of predictions. By addressing these challenges, the potential of AI-driven financial applications to deliver timely, personalized, and secure financial recommendations can be fully realized, advancing both user experience and the financial services sector.

REFERENCES

- [1] M. J. Zaki and W. Meira Jr, *Data mining and analysis: fundamental concepts and algorithms*. Cambridge University Press, 2014.
 - [2] I. Goodfellow, Y. Bengio, and A. Courville, *Deep learning*. MIT Press, 2016.
 - [3] F. Chollet, *Deep learning with Python*. Manning Publications, 2017.
 - [4] J. Heaton, N. Polson, and J. Witte, "Deep learning for finance: deep portfolios," *Applied Stochastic Models in Business and Industry*, vol. 33, no. 1, pp. 3–12, 2017.
 - [5] D. M. Nelson, A. C. Pereira, and R. A. de Oliveira, "Stock market's price movement prediction with lstm neural networks," in *IEEE International Joint Conference on Neural Networks (IJCNN)*. IEEE, 2017, pp. 1419–1426.
 - [6] M. Rezaei, M. R. F. Sani, and S. M. Homayouni, "Deep learning-based stock market prediction," *IEEE Access*, vol. 8, pp. 124 408–124 422, 2020.
 - [7] A. Esteva, B. Kuprel, R. A. Novoa, J. Ko, S. M. Swetter, H. M. Blau, and S. Thrun, "Dermatologist-level classification of skin cancer with deep neural networks," *Nature*, vol. 542, no. 7639, pp. 115–118, 2017.
 - [8] A. Agrawal, J. S. Gans, and A. Goldfarb, "Prediction machines: the simple economics of artificial intelligence," *Harvard Business Review*, 2018.
 - [9] J. Bollen, H. Mao, and X. Zeng, "Twitter's mood predicts the stock market," *Journal of Computational Science*, vol. 2, no. 1, pp. 1–8, 2011.
 - [10] V. Kumar and D. Shah, "Building and sustaining profitable customer loyalty for the 21st century," *Journal of Retailing*, vol. 80, no. 4, pp. 317–329, 2004.
 - [11] S. Rasp, P. D. Dueben, S. Scher, J. A. Weyn, S. Mouatadid, and N. Thuerey, "Weatherbench: a benchmark dataset for data-driven weather forecasting," *Quarterly Journal of the Royal Meteorological Society*, vol. 146, no. 730, pp. 370–385, 2020.
 - [12] R. Carbonneau, K. Laframboise, and R. Vahidov, "Application of machine learning techniques for supply chain demand forecasting," *European Journal of Operational Research*, vol. 184, no. 3, pp. 1140–1154, 2008.
 - [13] T. Young, D. Hazarika, S. Poria, and E. Cambria, "Recent trends in deep learning-based natural language processing," *IEEE Computational Intelligence Magazine*, vol. 13, no. 3, pp. 55–75, 2018.
-

-
- [14] K. Sun, "Fine-tuned bert models for sentiment analysis in financial marketplaces," *Journal of AI Applications in Economics*, vol. 13, no. 3, pp. 19–37, 2023.
- [15] S. Patel and R. Kumar, "Predictive analytics for budget planning in fintech," *Digital Finance Advances*, vol. 11, no. 2, pp. 67–81, 2022.
- [16] J. Lee, "Ai-driven fraud detection in digital payments," *International Journal of Financial Security*, vol. 8, no. 3, pp. 23–40, 2023.
- [17] V. Ramachandran, "Blockchain-enhanced ai for transparent financial systems," *Journal of Blockchain and AI*, vol. 10, no. 1, pp. 45–62, 2024.
- [18] Y. Chen, "Sentiment analysis in mobile banking: enhancing user experience through ai," *Journal of Financial Technology*, vol. 12, no. 1, pp. 34–49, 2024.
- [19] R. Gupta, "Integrating sentiment analysis with transaction data for merchant recommendations," *Fintech Innovations Quarterly*, vol. 9, no. 4, pp. 78–92, 2023.
- [20] H. Iqbal, "Combining tcn and n-beats models for financial forecasting," *AI and Finance Journal*, vol. 15, no. 3, pp. 56–71, 2022.
- [21] P. Johnson, "Knowledge graphs and sentiment analysis for decision support in fintech," *Advanced Computational Finance*, vol. 7, no. 2, pp. 88–104, 2024.
- [22] T. Sharma, "Challenges in deploying ai in fintech: a scalability perspective," *Journal of AI Ethics in Finance*, vol. 5, no. 2, pp. 101–115, 2023.
- [23] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: a survey," *ACM Computing Surveys*, vol. 41, no. 3, pp. 1–58, 2009.
- [24] M. El Abaji and R. A. Haraty, "Enhancing bitcoin forecast accuracy by integrating ai, sentiment analysis, and financial models," *Journal of Computing and Digital Systems*, <http://dx.doi.org/10.12785/ijcds/1571026011>, vol. 17, no. 1, pp. 1–15, 2025.
- [25] R. A. Haraty and S. Sobeh, "Unveiling insights from unstructured wealth: A comparative analysis of clustering techniques on blockchain cryptocurrency data," *Journal of Advances in Computing and Engineering*, <http://dx.doi.org/10.21622/ACE.2024.04.1.698>, vol. 4, no. 1, pp. 1–43, 2024.
-